MATHEMATICS

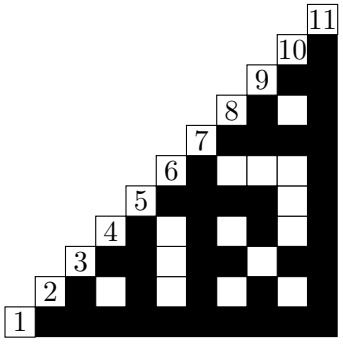
University *of* NORTH ALABAMA

Spring 2026 Math Problem of the Week: Problem #1

Due noon Tuesday, January 20
Open to all UNA undergraduates

Directions: Each Monday, a new problem will be posted on the bulletin board near the front desk and around the 3rd floor of CPTMTH. You have until noon of the proceeding Monday to deliver your written solution to me (Dr. Bryant) in CPTMTH 328 or, if you cannot find me, you may leave it with Ms. Buckley in CPTMTH 377. All work must be your own. You are required to justify your response fully. If you utilize a well-known theorem, state the theorem's full name and proposition, showing that the conditions of the theorem are met. Good luck!

Problem #1: The chart shown can be considered to extend to infinity. In it each square is blackened if and only if the numbers in the same row and column are relatively prime.



1. Where is the first black square with no edge or vertex adjacent to any other black square?
2. Where is the first white square with no edge or vertex adjacent to any other white square?
3. Where is the first white square with no edge or vertex adjacent to any black square?